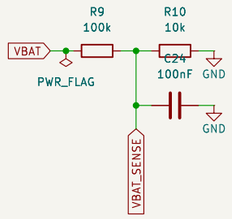
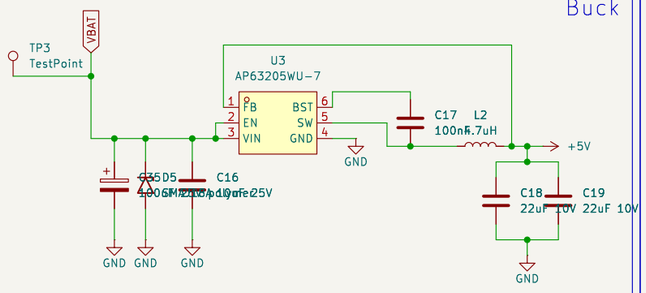
Power

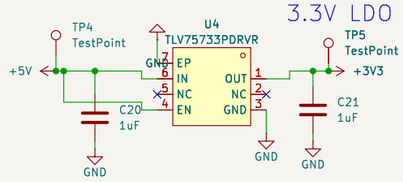
Battery sense divider



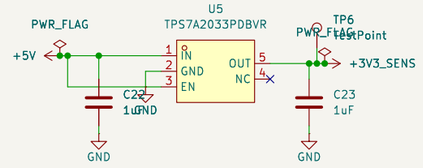
Buck



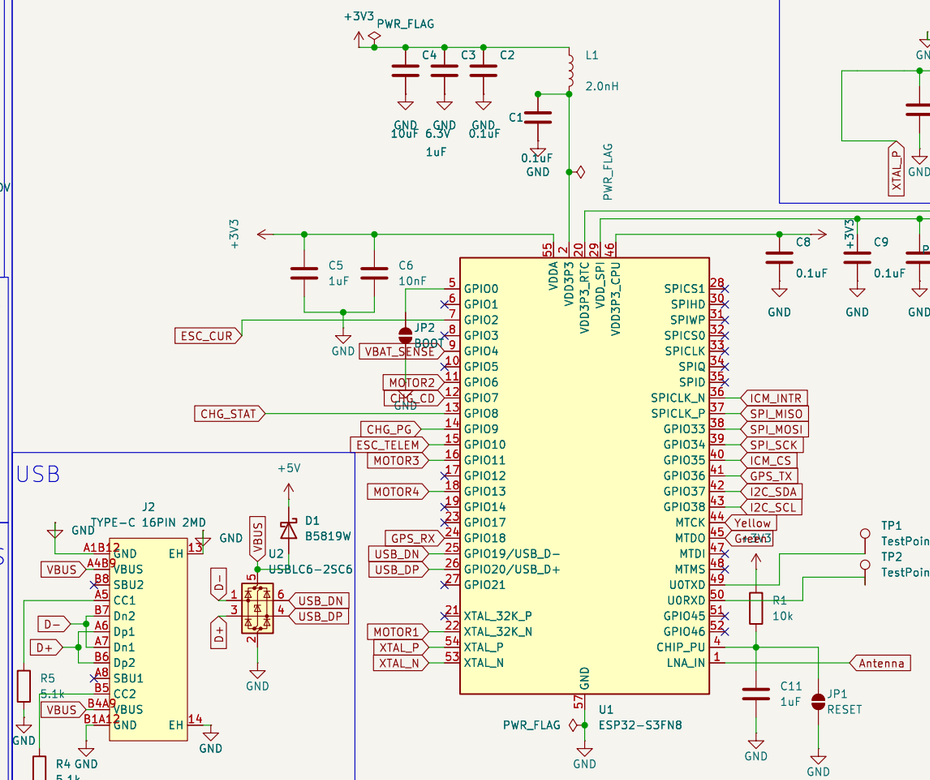
3.3V LDO – ESP, LEDs



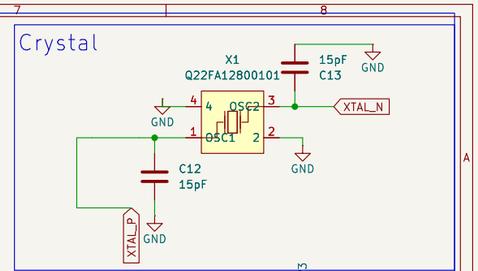
3.3V LDO IMU, BAROMETER, GPS



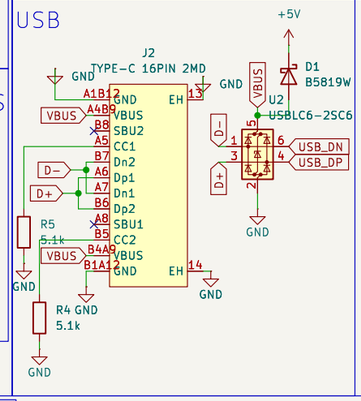
MCU



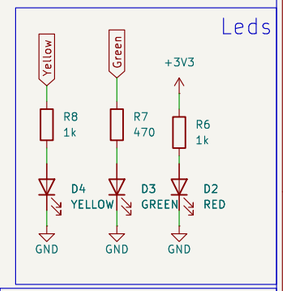
Crystal



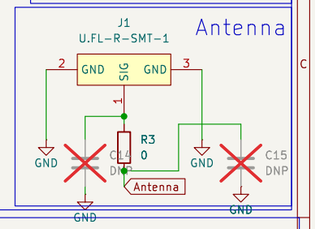
USB



Leds

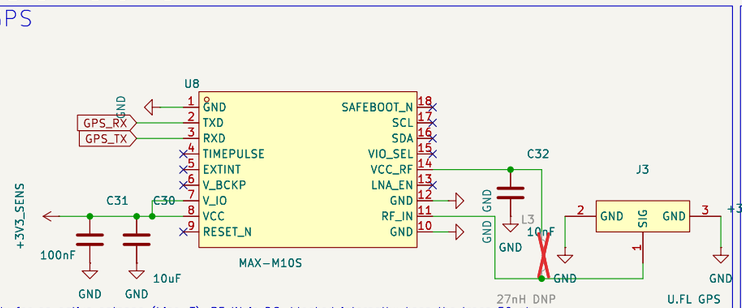


Antenna



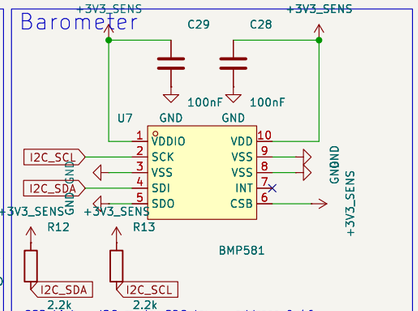
Sensors

GPS



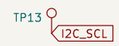
L3 only for an active antenna (bias-T). RF_IN is DC-blocked internally; keep the trace 50 ohm.

Barometer



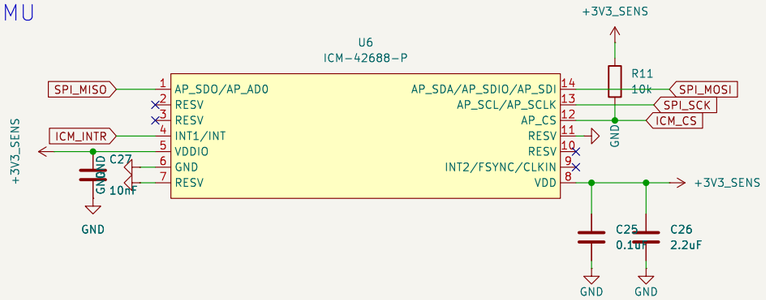
CSB high = I2C mode, SDO low = address 0x46

Test points



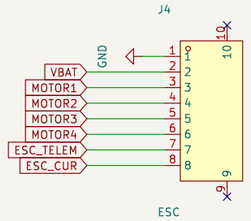
Bottom-side pads for probing the buses and recovering the ESP32.\nShort the RESET or BOOT pad pair to GND with tweezers.

IMU



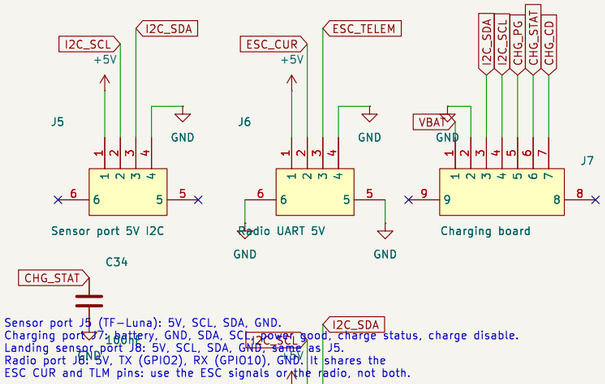
Connectors

ESC (JST-SH 8)

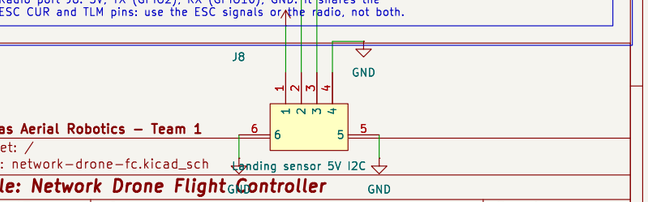


Pin order follows the Betaflight 4-in-1 standard:
GND, VBAT, CUR, TLM, M1-M4. Confirm against the ESC.

Ports (JST-GH 4)



Sensor port J5 (TF-Luna): 5V, SCL, SDA, GND.
Charging port J7: battery, GND, SDA, SCL, power good, charge status, charge disable.
Landing sensor port J8: 5V, SCL, SDA, GND, same as J5.
Radio port J6: 5V, TX (GPIO2), RX (GPIO10), GND. It shares the
ESC CUR and TLM pins: use the ESC signals or the radio, not both.



Texas Aerial Robotics – Team 1

Sheet: /
File: network-drone-fc.kicad_sch

Title: Network Drone Flight Controller

Size: A3	Date:	**Rev: A**
KiCad E.D.A. 9.0.7		Id: 1/1